

99.500 Applied Mathematics for Engineering Spring Semester AY2026

SOV & Laplace Transform

Week 10 Outline

- Separation of variables with non-homogeneous PDE and BCs
- Laplace transform for ODE

SOV Example 1

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

$$IC : C(x, 0) = 0$$

$$BC1 : C(0, t) = C_s$$

$$BC2 : C(L, t) = 0$$

$C(x, t) = C_{ss}(x) + C_t(x, t)$	
$C_{ss}(x)$	$C_t(x, t)$
$0 = \frac{d^2 C_{ss}}{dx^2}$	$\frac{\partial C_t}{\partial t} = D \frac{\partial^2 C_t}{\partial x^2}$
$\begin{cases} C_{ss}(0) = C_s \\ C_{ss}(L) = 0 \end{cases}$	$\begin{cases} IC : C_t(x, 0) = -C_{ss}(x) \\ BC1 : C_t(0, t) = 0 \\ BC2 : C_t(L, t) = 0 \end{cases}$

SOV Example 1

For the C_{ss} Part:

$$\begin{aligned}0 &= \frac{d^2 C_{ss}}{dx^2} &\implies C_{ss}(x) &= Ax + B \\C_{ss}(0) &= C_s &\implies B &= C_s \\C_{ss}(L) &= 0 &\implies A &= \frac{-C_s}{L} \\&&\implies C_{ss}(x) &= C_s \left(1 - \frac{x}{L}\right)\end{aligned}$$

For the C_t Part: $C_t(x, t) = X(x)T(t)$ $\implies \frac{T'}{DT} = \frac{X''}{X} = -\lambda^2$

T part: $T(t) = c_1 e^{-\lambda^2 Dt}$

X part: $X'' + \lambda^2 X = 0 \implies X = c_2 \cos(\lambda x) + c_3 \sin(\lambda x)$

BC1: $X(0) = 0 \implies c_2 = 0$

BC2: $X(L) = 0 \implies c_3 \sin(\lambda L) = 0 \implies \lambda = \frac{n\pi}{L}, n = 1, 2, 3, \dots$

SOV Example 1

$$C_t(x, t) = \sum_{n=1}^{\infty} X_n(x) T_n(t) = \sum_{n=1}^{\infty} A_n \sin(\lambda_n x) e^{-\lambda_n^2 D t} \text{ where } \lambda_n = \frac{n\pi}{L}, n = 1, 2, 3, \dots$$

Apply IC to determine A_n :

$$C_t(x, 0) = \sum_{n=1}^{\infty} A_n \sin(\lambda_n x) = -C_{ss}(x) = -C_s \left(1 - \frac{x}{L}\right)$$

$$A_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx} = \frac{\int_0^L -C_s \left(1 - \frac{x}{L}\right) \cdot \sin(\lambda_n x) \cdot 1 dx}{\int_0^L \sin^2(\lambda_n x) \cdot 1 dx} = \frac{-C_s \left(\frac{1}{\lambda_n}\right)}{\frac{L}{2}} = -\frac{2C_s}{n\pi}$$

SOV Example 2

$$\frac{\partial C}{\partial t} = \frac{D}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C}{\partial r} \right) - A - BC$$

$$IC : C(r, 0) = C_0$$

$$BC1 : \left. \frac{\partial C}{\partial r} \right|_{r=0} = 0$$

$$BC2 : C(r_0, t) = C_0$$

$C(r, t) = C_{ss}(r) + C_t(r, t)$	
$C_{ss}(r)$	$C_t(r, t)$
$0 = \frac{D}{r} \frac{d}{dr} \left(r \frac{dC_{ss}}{dr} \right) - A - BC_{ss}$	$\frac{\partial C_t}{\partial t} = \frac{D}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C_t}{\partial r} \right) - BC_t$
$\begin{cases} \left. \frac{dC_{ss}}{dr} \right _{r=0} = 0 \\ C_{ss}(r_0) = C_0 \end{cases}$	$\begin{cases} IC : C_t(r, 0) = C_0 - C_{ss}(r) \\ BC1 : \left. \frac{\partial C_t}{\partial r} \right _{r=0} = 0 \\ BC2 : C_t(r_0, t) = 0 \end{cases}$

SOV Example 2

For the C_t Part: $C_t(r, t) = R(r)T(t) \implies \frac{T'}{DT} + \frac{B}{D} = \frac{1}{Rr} \frac{d}{dr} \left(r \frac{dR}{dr} \right) = -\lambda^2$

T part: $\frac{T'}{DT} = -\lambda^2 - \frac{B}{D} \implies c_1 e^{-(\lambda^2 D + B)t}$

R part: $\frac{d}{dr} \left(r \frac{dR}{dr} \right) + \lambda^2 Rr = 0 \implies R = c_2 J_0(\lambda r) + c_3 Y_0(\lambda r)$

BC1 : $R'(0) = 0 \implies c_3 = 0$

BC2 : $R(r_0) = 0 \implies c_2 J_0(\lambda r_0) = 0$

$\implies \lambda_n r_0$ is the n-th root of $J_0(x) = 0$

$$C_t(r, t) = \sum_{n=1}^{\infty} R_n(r) T_n(t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-(\lambda^2 D + B)t}$$

$$A_n = \frac{\int_0^{r_0} (C_0 - C_{ss}) J_0(\lambda_n r) r dr}{\int_0^{r_0} [J_0(\lambda_n r)]^2 r dr}$$

SOV Example 3

$$\nabla^2 T = 0 \quad \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \right)$$

$$BC_{x1} : \left. \frac{\partial T}{\partial x} \right|_{x=0} = 0$$

$$BC_{x2} : T(L, y) = 0$$

$$BC_{y1} : T(x, 0) = f(x)$$

$$BC_{y2} : T(x, H) = 0$$

$$T(x, y) = X(x)Y(y) \implies \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2 \quad \text{Why?}$$

$$X'' + \lambda^2 X = 0 \quad \text{with} \quad \begin{cases} X'(0) = 0 \\ X(L) = 0 \end{cases} \implies X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$
$$X'(0) = 0 \implies B = 0$$
$$X(L) = 0 \implies \cos(\lambda L) = 0$$
$$\implies \lambda_n L = \left(n - \frac{1}{2} \right) \pi, \quad n = 1, 2, 3 \dots$$

SOV Example 3

$$Y_n'' - \lambda_n^2 Y_n = 0 \implies Y_n(y) = A_n \cosh(\lambda_n y) + B_n \sinh(\lambda_n y)$$

$$Y_n(H) = 0 \implies B_n = \frac{-A_n}{\tanh(\lambda_n H)}$$

$$\implies Y_n(y) = A_n \frac{\sinh[\lambda_n(H - y)]}{\sinh(\lambda_n H)}$$

$$T(x, y) = \sum_{n=1}^{\infty} X_n(x) Y_n(y) = \sum_{n=1}^{\infty} A_n \cos(\lambda_n x) \frac{\sinh[\lambda_n(H - y)]}{\sinh(\lambda_n H)}$$

Apply BC_{y1} to determine A_n :

$$T(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \cos(\lambda_n x)$$

$$A_n = \frac{\int_0^L f(x) \cos(\lambda_n x) dx}{\int_0^L \cos^2(\lambda_n x) dx} \text{ with } \lambda_n = \left(n - \frac{1}{2}\right) \frac{\pi}{L}$$

SOV Example 4

$$\nabla^2 T = 0 \quad \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \right)$$

$$BC_{x1} : \left. \frac{\partial T}{\partial x} \right|_{x=0} = 0$$

$$BC_{x2} : T(L, y) = f_2(y)$$

$$BC_{y1} : T(x, 0) = f_1(x)$$

$$BC_{y2} : T(x, H) = 0$$

$$T(x, y) = T_1(x, y) + T_2(x, y)$$

	$\frac{\partial^2 T_1}{\partial x^2} + \frac{\partial^2 T_1}{\partial y^2} = 0$	$\frac{\partial^2 T_2}{\partial x^2} + \frac{\partial^2 T_2}{\partial y^2} = 0$
	$T_1(x, y)$	$T_2(x, y)$
$x = 0$	$\frac{\partial T_1}{\partial x} = 0$	$\frac{\partial T_2}{\partial x} = 0$
$x = L$	$T_1 = 0$	$T_2 = f_2(y)$
$y = 0$	$T_1 = f_1(x)$	$T_2 = 0$
$y = H$	$T_1 = 0$	$T_2 = 0$
	Already solved	Solve for T_2

SOV Example 4

$$T_2(x, y) = X(x)Y(y) \implies \frac{X''}{X} = -\frac{Y''}{Y} = \lambda^2$$

$$Y'' = -\lambda_n^2 Y \text{ with } \begin{cases} Y(0) = 0 \\ Y(H) = 0 \end{cases} \implies Y(y) = A_n \cos(\lambda_n y) + B_n \sin(\lambda_n y)$$

$$Y(0) = 0 \implies A_n = 0$$

$$Y(H) = 0 \implies \sin(\lambda_n H) = 0, \lambda_n H = n\pi, n = 1, 2, 3, \dots$$

$$X'' = \lambda^2 X \text{ with } \begin{cases} X'(0) = 0 \\ X(L) = ?? \end{cases} \implies X_n(x) = C_n \cosh(\lambda_n x) + D_n \sinh(\lambda_n x)$$
$$X'(0) = 0 \implies D_n = 0$$

$$T_2(x, y) = \sum_{n=1}^{\infty} E_n \cosh(\lambda_n x) \sin(\lambda_n y)$$

$$T_2(L, y) = \sum_{n=1}^{\infty} E_n \cosh(\lambda_n L) \sin(\lambda_n y) = f_2(y)$$

$$E_n \cosh(\lambda_n L) = \frac{\int_0^H f_2(y) \sin(\lambda_n y) dy}{\int_0^H \sin^2(\lambda_n y) dy}$$

Laplace Transform

$$\mathcal{L}[f(t)] \equiv \bar{f}(s) = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}^{-1}[\bar{f}(s)] \equiv f(t) = \frac{1}{2\pi i} \lim_{\omega \rightarrow \infty} \int_{\sigma_0 - i\omega}^{\sigma_0 + i\omega} e^{st} \bar{f}(s) ds$$

Existence Theorem for Laplace Transform

If $f(t)$ is defined and piecewise continuous on every finite interval on the semi-axis $t \geq 0$ and satisfies

$$|f(t)| \leq M e^{kt}$$

for all $t \geq 0$ and some constants M and k , then the Laplace transform $\mathcal{L}(f)$ exists for all $s > k$.

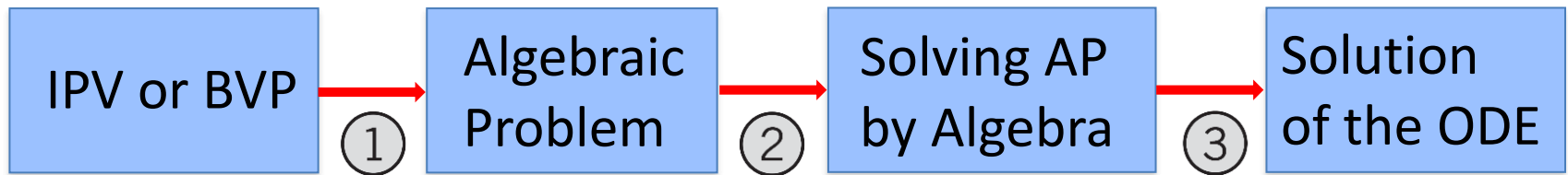
Some Properties of Laplace Transform

1. Linearity: $\mathcal{L}\{af(t) + bg(t)\} = a\bar{f}(s) + b\bar{g}(s)$
2. Differentiation of Function: $\mathcal{L}\{f^n(t)\} = s^n\bar{f}(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{n-1}(0)$
3. Integration of Function: $\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \frac{1}{s}\bar{f}(s)$
4. s-Shifting: $\mathcal{L}\{e^{at}f(t)\} = \bar{f}(s-a)$ and $\mathcal{L}^{-1}\{\bar{f}(s-a)\} = e^{at}f(t)$
5. Convolution: $\mathcal{L}\left\{\int_0^t f(t-u)g(u)du\right\} = \bar{f}(s)\bar{g}(s)$
6. Differentiation of Transform: $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n \bar{f}(s)}{ds^n}$
7. f Periodic with Period p : $\bar{f}(s) = \frac{1}{1 - e^{-ps}} \int_0^p e^{-st} f(t) dt$
8. Asymptotic small-time behavior: $\lim_{t \rightarrow 0} f(t) \sim \mathcal{L}^{-1}\left\{\lim_{s \rightarrow \infty} \bar{f}(s)\right\}$
9. Heaviside expansion: Let $P(s)$ and $Q(s)$ be polynomial in s where $P(s)$ has degree less than of $Q(s)$. If $Q(s)$ has n distinct roots α_k for $k = 1, 2, \dots, n$ that is

$$Q(s) = (s - \alpha_1)(s - \alpha_2) \cdots (s - \alpha_n)$$

$$\text{then } \mathcal{L}^{-1}\left\{\frac{P(s)}{Q(s)}\right\} = \sum_{k=1}^n \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t}$$

Solving ODE by Laplace Transform



1. The given ODE is transformed into an algebraic equation, called the subsidiary equation.
2. The subsidiary equation is solved by purely algebraic manipulations.
3. The solution in Step 2 is transformed back, resulting in the solution of the given problem.

LT Example 1

$$\frac{d^2 r}{dt^2} = -\lambda r$$

$$IC1 : r(0) = r_0$$

$$IC2 : r'(0) = 0$$

Laplace transform tables	
Time space	Laplace
$\cos at$	$\frac{s}{s^2 + a^2}$
$\sin at$	$\frac{a}{s^2 + a^2}$

$$\mathcal{L}\{\text{ODE}\} \implies s^2 \bar{r} - [r'(0) + sr(0)] = -\lambda \bar{r}$$

$$\implies s^2 \bar{r} - [0 + sr_0] = -\lambda \bar{r}$$

$$\implies \bar{r} = \frac{sr_0}{s^2 + \lambda}$$

$$\implies r(t) = \mathcal{L}^{-1} \left\{ \frac{sr_0}{s^2 + \lambda} \right\} = r_0 \cos(\sqrt{\lambda}t)$$

LT Example 2

$$\frac{d^2 r}{dt^2} = -k \frac{dr}{dt} + c \sin(\omega t)$$

$$IC1 : r(0) = 0$$

$$IC2 : r'(0) = 0$$

$$\begin{aligned}\mathcal{L}\{\text{ODE}\} &\implies s^2 \bar{r} - [r'(0) + sr(0)] = -k [s\bar{r} - r(0)] + \frac{c\omega}{s^2 + \omega^2} \\ &\implies \bar{r} = \frac{c\omega}{s(s+k)(s^2 + \omega^2)}\end{aligned}$$

How to invert the $\bar{r}$?

Using Convolution Property

$$\mathcal{L}^{-1} [\bar{f}(s)\bar{g}(s)] = \int_0^t f(t-u)g(u)du$$

$$\bar{r} = \frac{c\omega}{s(s+k)(s^2+\omega^2)} \implies \bar{r} = \frac{1}{s(s+k)} \frac{c\omega}{(s^2+\omega^2)} = \bar{g}(s)\bar{f}(s)$$

$$\bar{g}(s) = \frac{1}{s(s+k)} \implies g(t) = \frac{1}{k} (1 - e^{-kt})$$

$$\bar{f}(s) = \frac{c\omega}{(s^2+\omega^2)} \implies f(t) = c \sin(\omega t)$$

$$r(t) = \frac{c}{k} \int_0^t \sin [\omega(t-u)] (1 - e^{-ku}) du$$

Using Convolution Property

$$\begin{aligned}\sin \omega(t-u) &= \sin \omega t \cos \omega u - \cos \omega t \sin \omega u \\ \frac{r(t)k}{c} &= \int_0^t (\sin \omega t \cos \omega u - \cos \omega t \sin \omega u) (1 - e^{-ku}) du \\ &= \int_0^t (\sin \omega t \cos \omega u - \cos \omega t \sin \omega u) du \\ &\quad - \int_0^t e^{-ku} (\sin \omega t \cos \omega u - \cos \omega t \sin \omega u) du \\ r(t) &= \frac{c}{k\omega} - \frac{c\omega e^{-kt}}{k(\omega^2 + k^2)} - \frac{c}{\omega(\omega^2 + k^2)}(k \cos \omega t + \omega \sin \omega t)\end{aligned}$$

Heaviside Expansion

$$\bar{r} = \frac{1}{s(s+k)} \frac{c\omega}{(s^2 + \omega^2)} = \frac{P(s)}{Q(s)} = \frac{1}{s(s+k)} \frac{c\omega}{(s+i\omega)(s-i\omega)}$$

$$P(s) = c\omega \quad Q(s) = s(s+k)(s+i\omega)(s-i\omega)$$

To use the Heaviside expansion method for partial fraction decomposition in Laplace Transform, the following three criteria must be met:

1. $P(s)$ and $Q(s)$ must be polynomials of s .
2. The degree of $P(s)$ must be lower than the degree of $Q(s)$ (i.e., $\deg P(s) < \deg Q(s)$).
3. The roots of $Q(s) = 0$ must be distinct (i.e., no repeated roots).

$$Q(s) = 0 \implies \alpha_1 = 0, \alpha_2 = -k, \alpha_3 = -i\omega, \alpha_4 = i\omega$$

$$\mathcal{L}^{-1} \{ \bar{r} \} = \mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} = \sum_{k=1}^N \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t}$$

Heaviside Expansion

$$P(s) = c\omega \implies P(\alpha_k) = c\omega \text{ (for any } k)$$

$$Q(s) = s(s+k)(s^2 + \omega^2)$$

$$\begin{aligned} Q'(s) &= s(s+k)(2s) + (s^2 + \omega^2)[s(s+k)]' \\ &= 2s^2(s+k) + (s^2 + \omega^2)[(s+k) + s] \\ &= (3s^2 + \omega^2)(s+k) + s(s^2 + \omega^2) \end{aligned}$$

$$Q'(\alpha_1) = k\omega^2$$

$$Q'(\alpha_2) = -k(\omega^2 + k^2)$$

$$Q'(\alpha_3) = -2\omega^2(k - i\omega)$$

$$Q'(\alpha_4) = -2\omega^2(k + i\omega)$$

Heaviside Expansion

$$\begin{aligned}r(t) &= \sum_{k=1}^N \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t} \\&= \frac{c\omega}{k\omega^2} - \frac{c\omega e^{-kt}}{k(\omega^2 + k^2)} - \frac{c\omega e^{-i\omega t}}{2\omega^2(k - i\omega)} - \frac{c\omega e^{i\omega t}}{2\omega^2(k + i\omega)} \\&= \frac{c\omega}{k\omega^2} - \frac{c\omega e^{-kt}}{k(\omega^2 + k^2)} - c\omega \frac{(k + i\omega)e^{-i\omega t} + (k - i\omega)e^{i\omega t}}{2\omega^2(k^2 + \omega^2)} \\&= \frac{c\omega}{k\omega^2} - \frac{c\omega e^{-kt}}{k(\omega^2 + k^2)} - \frac{c(k \cos \omega t + \omega \sin \omega t)}{\omega(k^2 + \omega^2)}\end{aligned}$$

Announcements

Week 10 Homework

- Due Date: Week 11 class – 2:00 PM, April 8, 2026
- Given 3 questions, only required to do **2** questions

Mid-Term Project

- Final submission of all the files (“exam report”, ppt slides, codes) by 23:59, **April 3**, 2026